



US012406882B2

(12) **United States Patent**
Lee et al.

(10) **Patent No.:** **US 12,406,882 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SEMICONDUCTOR ELEMENT AND
METHOD FOR MANUFACTURING THE
SAME**

USPC 257/751
See application file for complete search history.

(71) Applicant: **MACRONIX INTERNATIONAL
CO., LTD.**, Hsinchu (TW)

(72) Inventors: **Dai-Ying Lee**, Hsinchu County (TW);
Yu-Chao Huang, Hsinchu (TW)

(73) Assignee: **MACRONIX INTERNATIONAL
CO., LTD.**, Hsinchu (TW)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 522 days.

(21) Appl. No.: **17/821,479**

(22) Filed: **Aug. 23, 2022**

(65) **Prior Publication Data**

US 2024/0071821 A1 Feb. 29, 2024

(51) **Int. Cl.**
H01L 21/768 (2006.01)
H01L 23/522 (2006.01)
H01L 23/532 (2006.01)

(52) **U.S. Cl.**
CPC .. **H01L 21/76843** (2013.01); **H01L 21/76877**
(2013.01); **H01L 23/5226** (2013.01); **H01L**
23/53266 (2013.01)

(58) **Field of Classification Search**
CPC H01L 21/76843; H01L 21/76877; H01L
23/5226; H01L 23/53266

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Primary Examiner — Tu-Tu V Ho

(74) *Attorney, Agent, or Firm* — McClure, Qualey &
Rodack, LLP

(57) **ABSTRACT**

A semiconductor element and a method for manufacturing the same are provided. The semiconductor element includes a plug and a via on the plug and electrically connected to the plug. The plug includes a tungsten plug and a conductive layer on the tungsten plug. The tungsten plug and the conductive layer include different materials. The tungsten plug has a first width in a lateral direction. The conductive layer has a second width in the lateral direction. The second width is greater than or equal to the first width. The conductive layer is between the via and the tungsten plug.

20 Claims, 9 Drawing Sheets

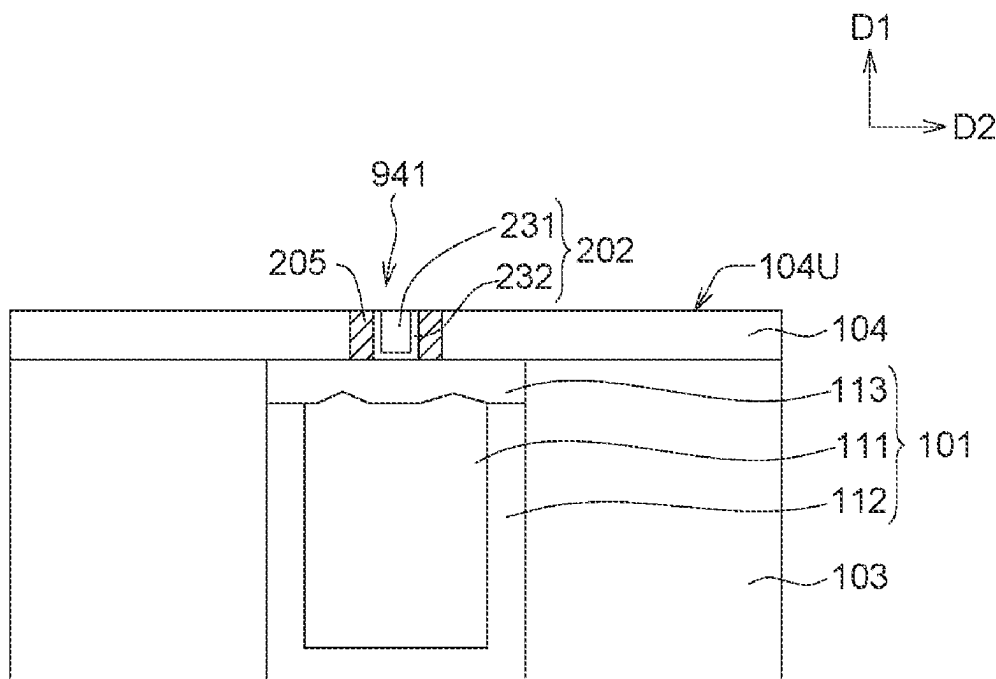


FIG. 2

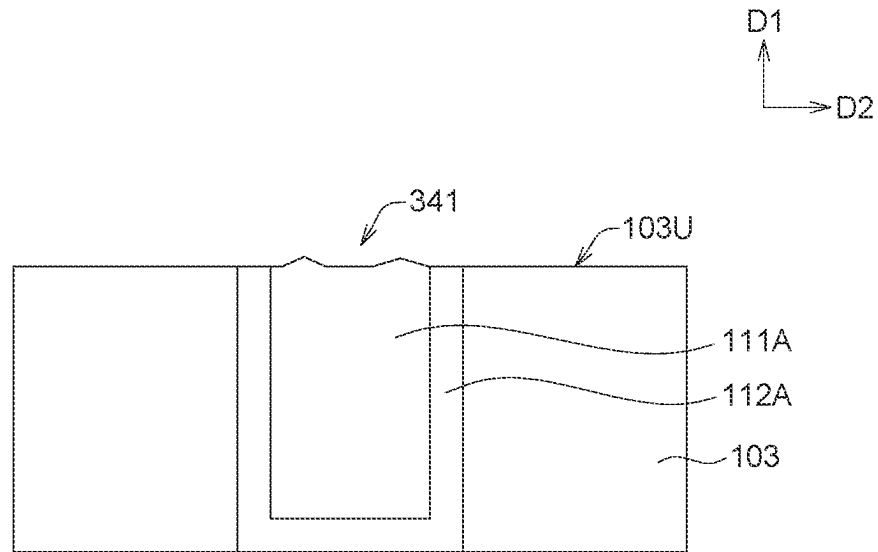


FIG. 3

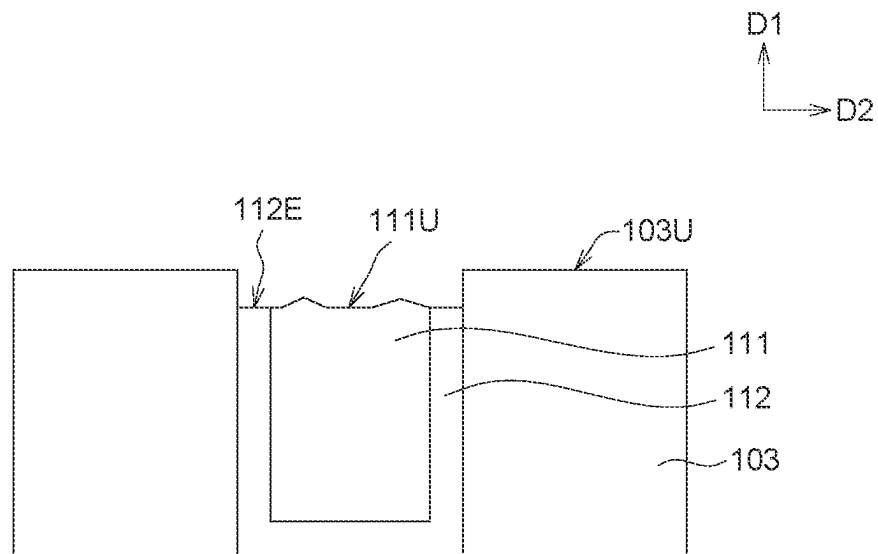


FIG. 4

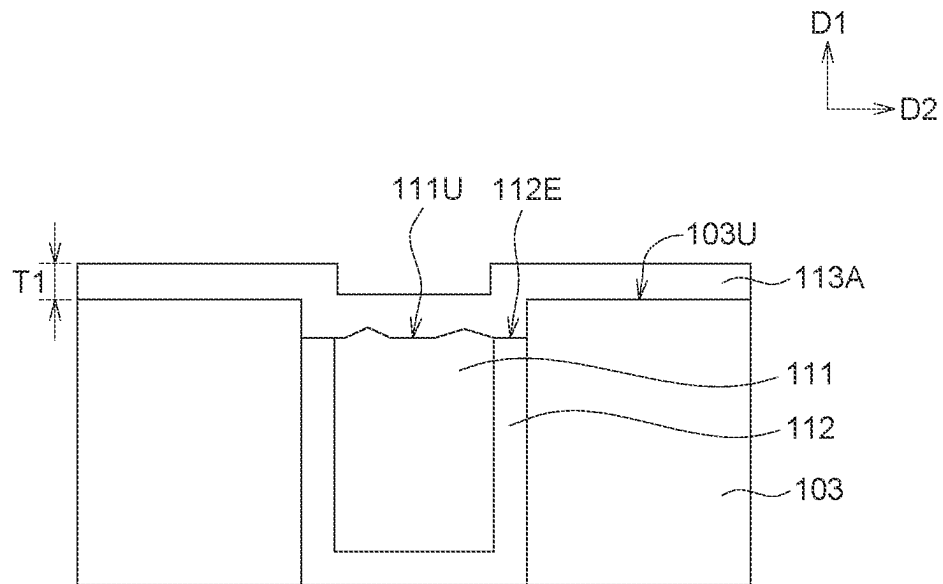


FIG. 5

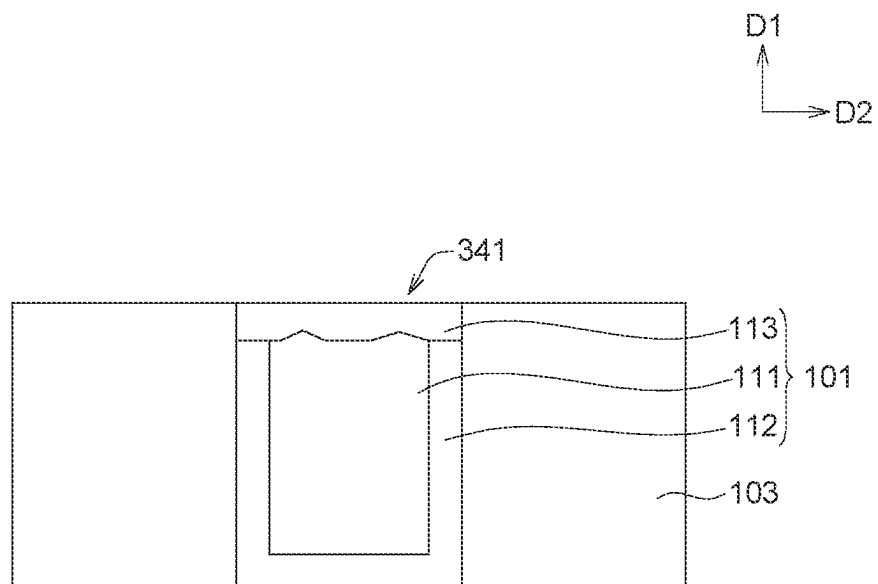


FIG. 6

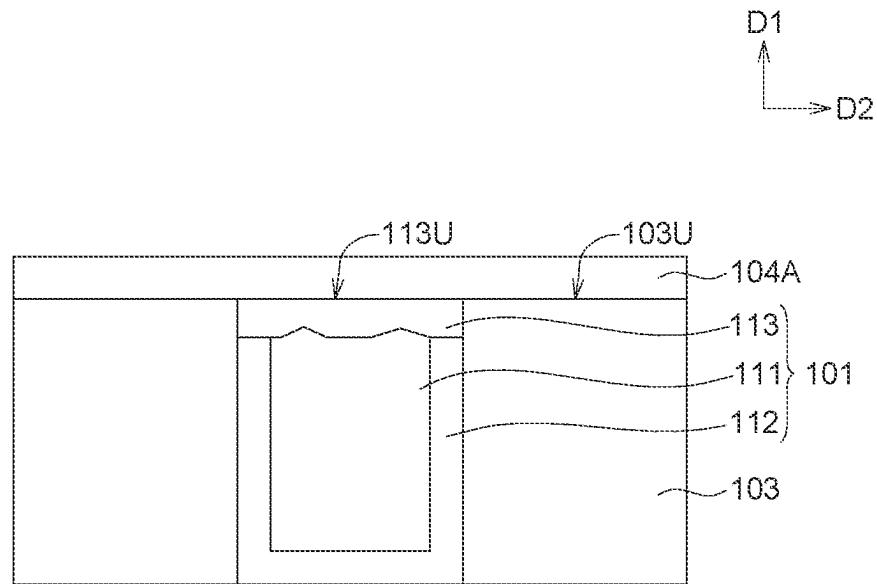


FIG. 7

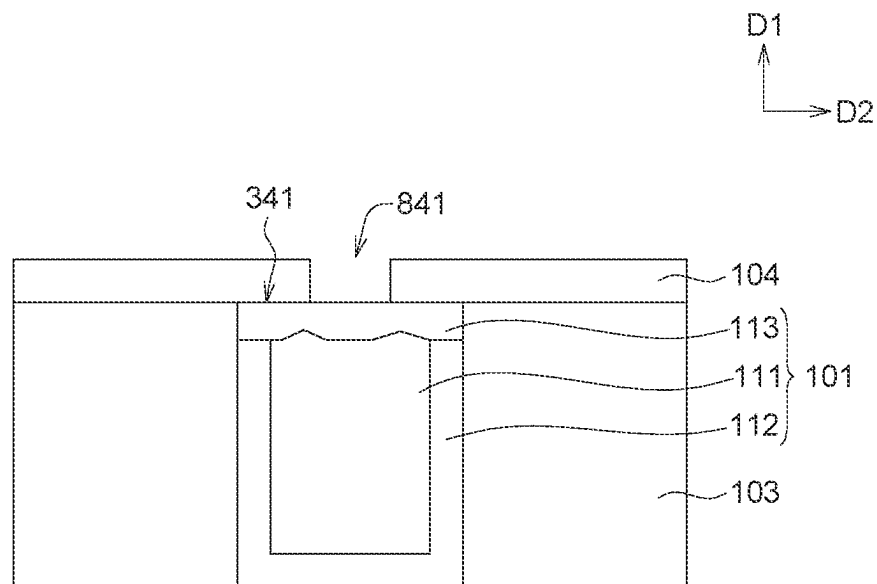


FIG. 8

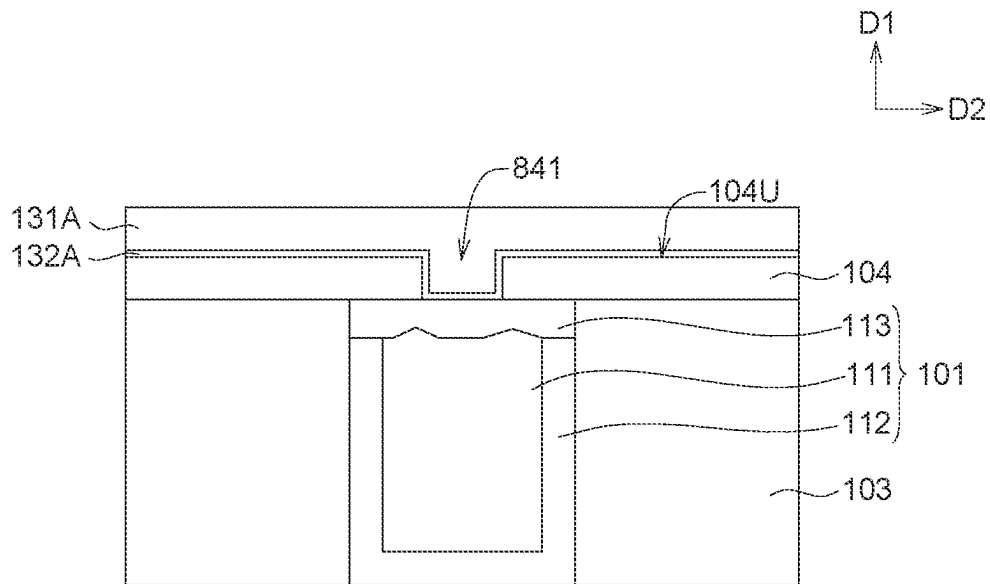


FIG. 9

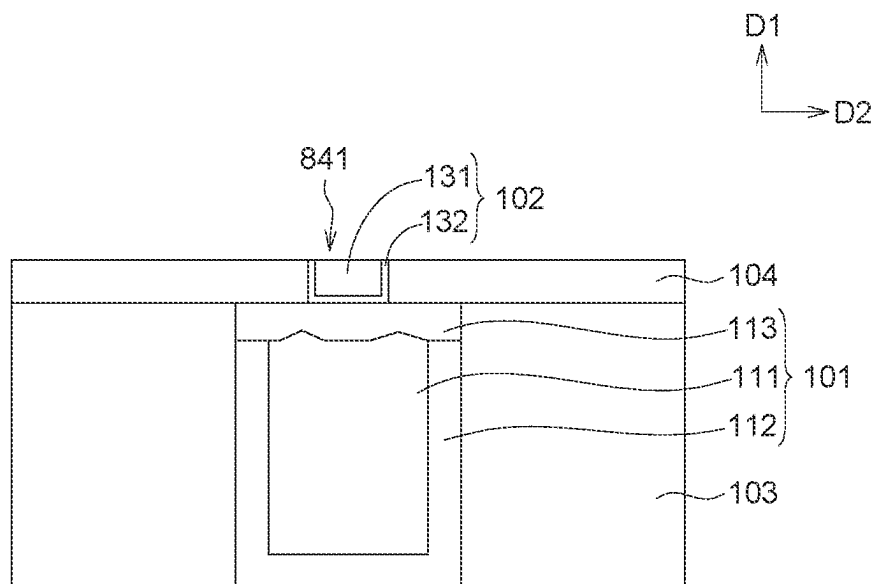


FIG. 10

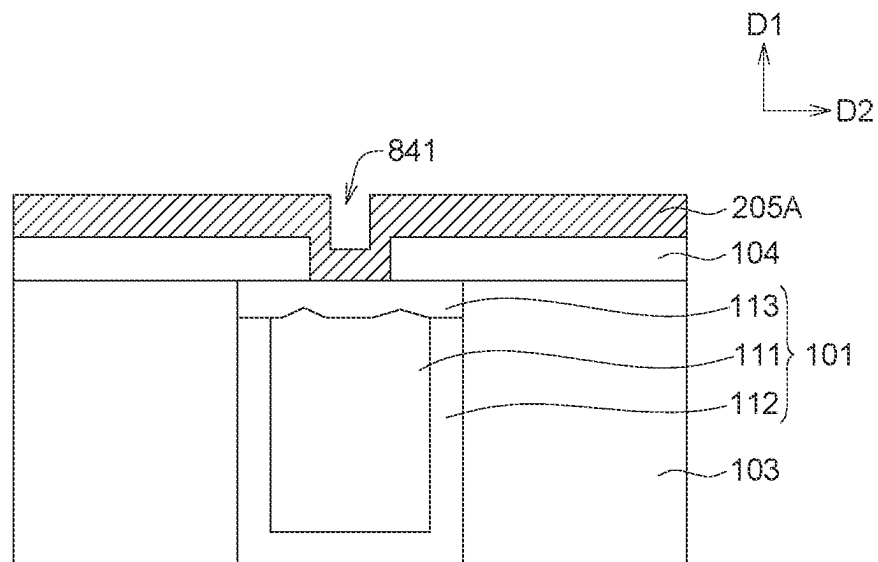


FIG. 11

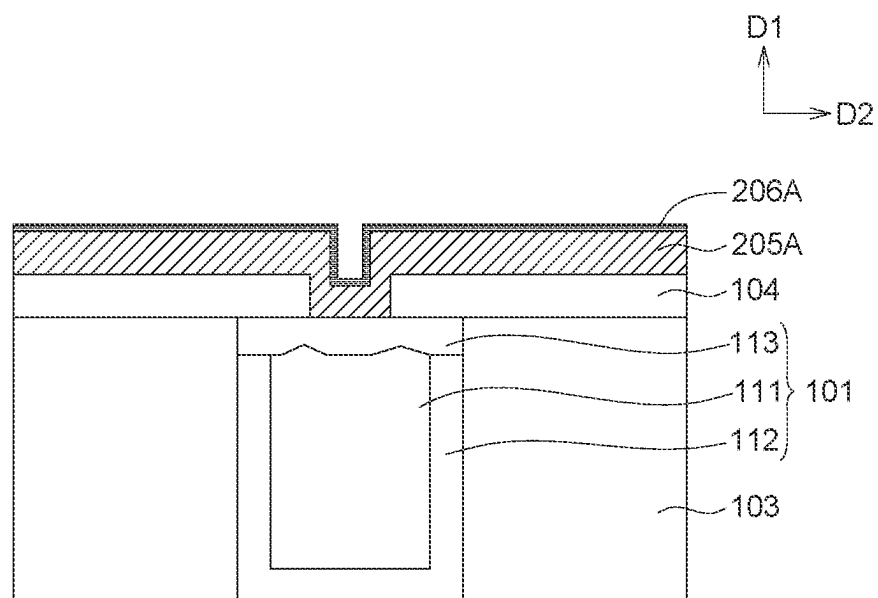


FIG. 12

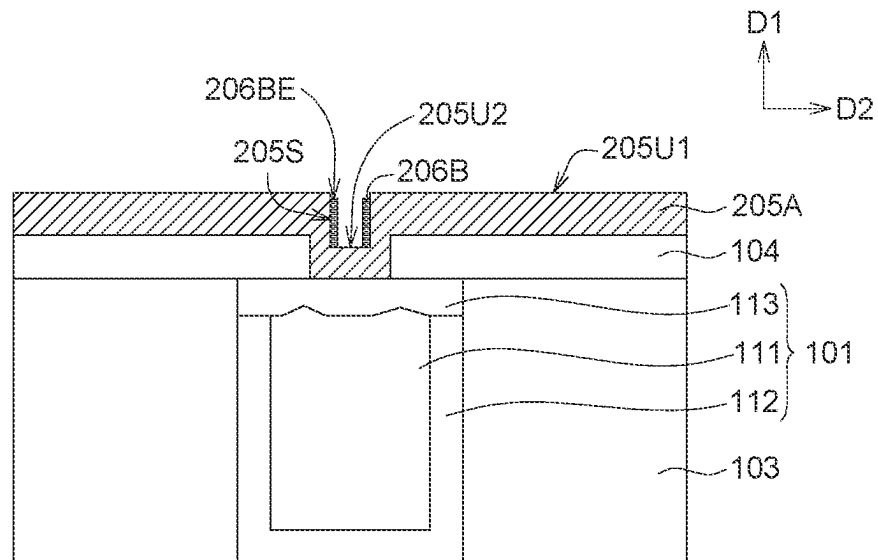


FIG. 13

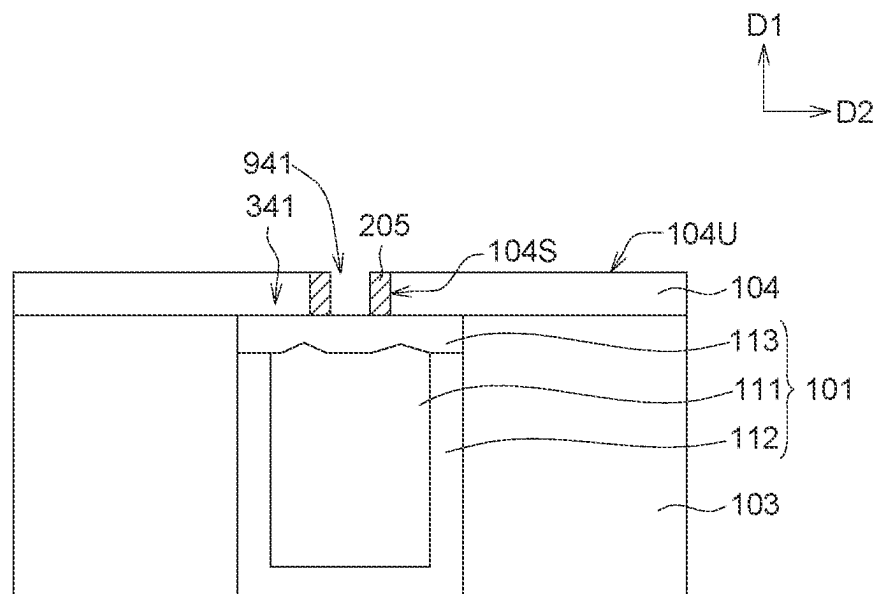


FIG. 14

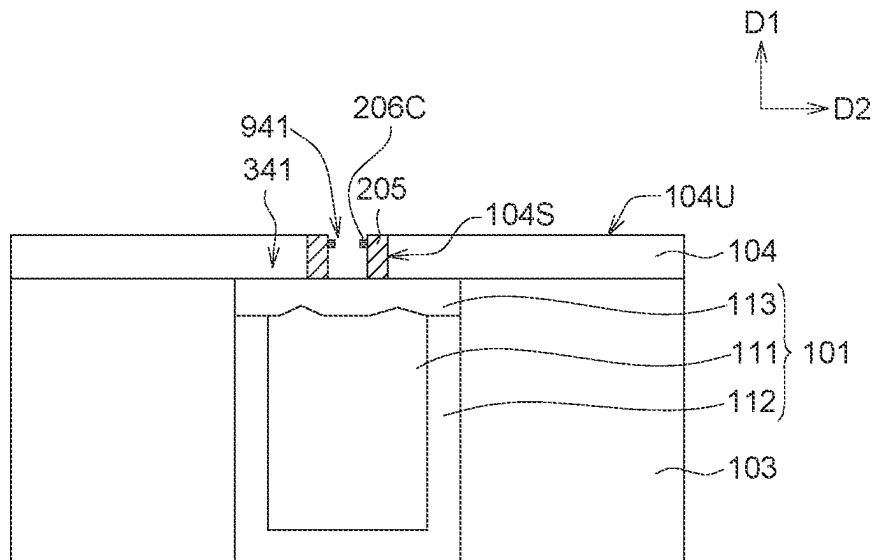


FIG. 14A

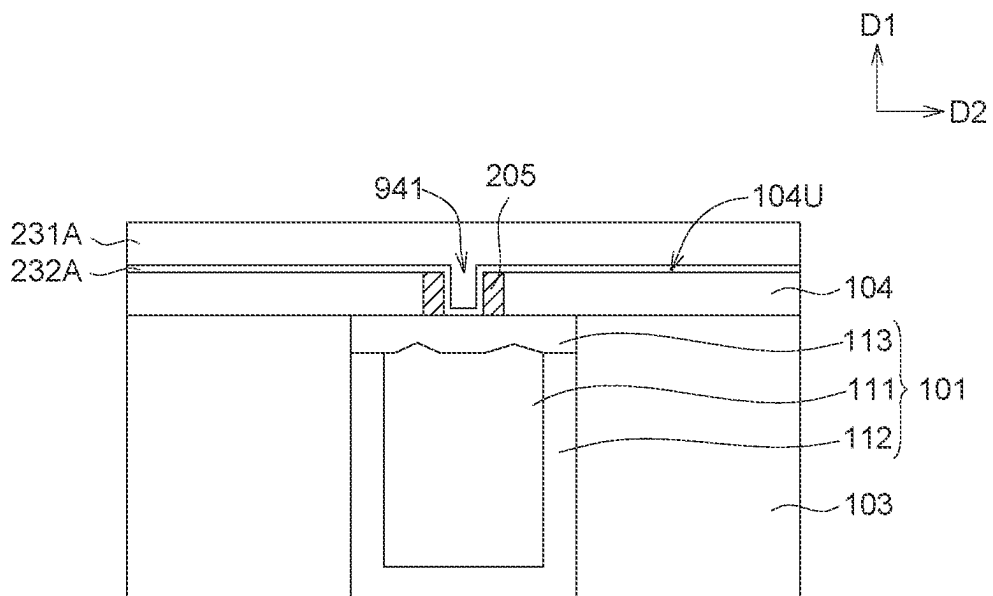


FIG. 15

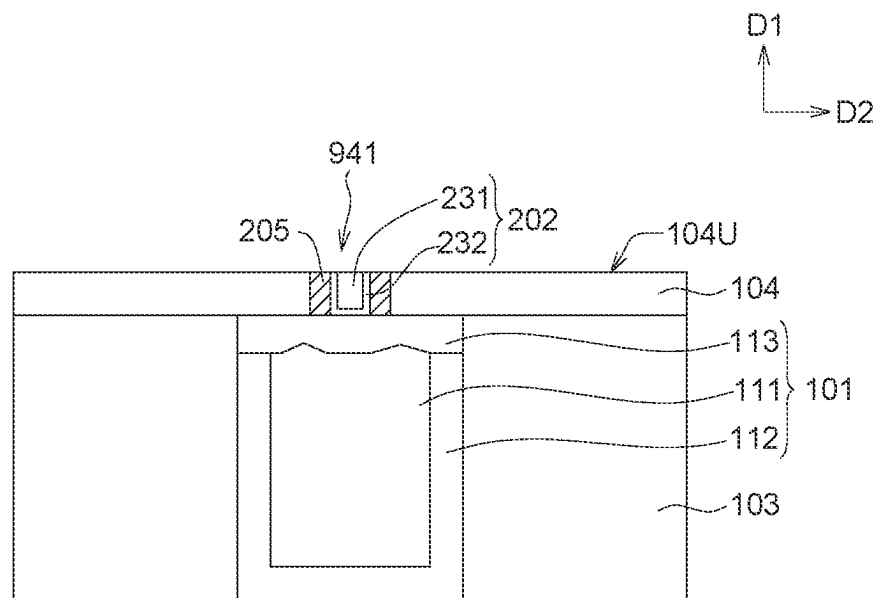


FIG. 16

1

SEMICONDUCTOR ELEMENT AND METHOD FOR MANUFACTURING THE SAME

BACKGROUND

Technical Field

The present disclosure relates to a semiconductor element and method for manufacturing the same.

Description of the Related Art

As the size of semiconductor devices shrinks and the integration increases, it becomes increasingly difficult to control the profiles of elements in the semiconductor device, which may affect the electrical performance of the semiconductor device. For example, the profile defects of elements may cause elements not to be formed in the expected positions, resulting in problems such as open circuits and short circuits.

It is important to provide technology for semiconductor elements with improved electrical performance.

SUMMARY

The present disclosure relates to a semiconductor element and a method for manufacturing the same.

According to an embodiment of the present disclosure, a semiconductor element is provided. The semiconductor element includes a plug and a via on the plug and electrically connected to the plug. The plug includes a tungsten plug and a conductive layer on the tungsten plug. The tungsten plug and the conductive layer include different materials. The tungsten plug has a first width in a lateral direction. The conductive layer has a second width in the lateral direction. The second width is greater than or equal to the first width. The conductive layer is between the via and the tungsten plug.

According to another embodiment of the present disclosure, a method for manufacturing the semiconductor element is provided. The method includes the following steps. Forming an opening in a dielectric layer. Forming a tungsten plug in the opening. Forming a conductive layer on the tungsten plug, wherein the conductive layer is in the opening. Forming a via on the conductive layer, wherein the via is electrically connected to the conductive layer and the tungsten plug, the tungsten plug and the conductive layer comprises different materials.

The above and other embodiments of the disclosure will become better understood with regard to the following detailed description of the non-limiting embodiment(s). The following description is made with reference to the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a cross-sectional view of a semiconductor element according to an embodiment of the present disclosure.

FIG. 2 illustrates a cross-sectional view of a semiconductor element according to another embodiment of the present disclosure.

FIGS. 3-10 schematically illustrate a method for manufacturing a semiconductor element according to an embodiment of the present disclosure.

2

FIGS. 11-16 schematically illustrate a method for manufacturing a semiconductor element according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

Various embodiments will be described more fully hereinafter with reference to accompanying drawings, which are provided for illustrative and explaining purposes rather than a limiting purpose. For clarity, the components may not be drawn to scale. In addition, some components and/or reference numerals may be omitted from some drawings. It is contemplated that the elements and features of one embodiment can be beneficially incorporated in another embodiment without further recitation.

Referring to FIG. 1. FIG. 1 illustrates a cross-sectional view of a semiconductor element 10 according to an embodiment of the present disclosure. The semiconductor element 10 includes a plug 101, a via 102 electrically connected to the plug 101, a dielectric layer 103 and a dielectric film 104. The plug 101 may be in the dielectric layer 103. The plug 101 includes a tungsten plug 111, a barrier layer 112 and a conductive layer 113. The barrier layer 112 may be on a sidewall 111S and a lower surface 111B of the tungsten plug 111. Part of the barrier layer 112 may be between the tungsten plug 111 and the dielectric layer 103. In an embodiment, the barrier layer 112 directly contacts the dielectric layer 103 and/or the tungsten plug 111. The conductive layer 113 is between the via 102 and the tungsten plug 111. The conductive layer 103 is on an upper surface 111U of the tungsten plug 111. In an embodiment, the conductive layer 113 covers the upper surface 111U of the tungsten plug 111 and an end portion 112E of the barrier layer 112. In an embodiment, the conductive layer 113 directly contacts the tungsten plug 111 and the barrier layer 112. The tungsten plug 111 and the conductive layer 113 may include different materials.

In an embodiment, an upper surface 113U of the conductive layer 113 and an upper surface 103U of the dielectric layer 103 may be coplanar. In other embodiments, the upper surface 113U of the conductive layer 113 is lower than or higher than the upper surface 103U of the dielectric layer 103 in a longitudinal direction D1. For example, in the embodiment in which the upper surface 113U of the conductive layer 113 is higher than the upper surface 103U of the dielectric layer 103 in the longitudinal direction D1, the absolute value of the height difference between the upper surface 113U of the conductive layer 113 and the upper surface 103U of the dielectric layer 103 may be greater than 0 and less than or equal to 300 Angstroms (Å). For example, in the embodiment in which the upper surface 113U of the conductive layer 113 is lower than the upper surface 103U of the dielectric layer 103 in the longitudinal direction D1, the absolute value of the height difference between the upper surface 113U of the conductive layer 113 and the upper surface 103U of the dielectric layer 103 may be greater than 0 and less than 300 Angstroms (Å). The upper surface 111U of the tungsten plug 111 may be lower than the upper surface 103U of the dielectric layer 103. A roughness of the upper surface 113U of the conductive layer 113 may be less than a roughness of the upper surface 111U of the tungsten plug 111.

The via 102 may be in the dielectric film 104. The via 102 is on the plug 101. The plug 101 and the via 102 may overlap with each other in the longitudinal direction D1. The via 102 includes a conductive via 131 and a barrier film 132. The barrier film 132 may be on a sidewall 131S and a lower

surface **131B** of the conductive via **131**. Part of the barrier film **132** may be between the conductive via **131** and the dielectric film **104**. Another part of the barrier film **132** may be between the conductive layer **113** and the conductive via **131**. In an embodiment, the barrier film **132** directly contacts the dielectric film **104** and/or the conductive via **131**. In an embodiment, the barrier film **132** directly contacts the conductive layer **113**. The conductive via **131** of the via **102**, the barrier film **132** of the via **102**, the conductive layer **113** of the plug **101**, the tungsten plug **111** of the plug **101** and the barrier layer **112** of the plug **101** are electrically connected to each other.

The tungsten plug **111** has a first width **W1** in a lateral direction **D2**. The conductive layer **113** has a second width **W2** in the lateral direction **D2**. The second width **W2** of the conductive layer **113** may be greater than or equal to the first width **W1** of the tungsten plug **111**. In an embodiment in which the second width **W2** of the conductive layer **113** is equal to the first width **W1** of the tungsten plug **111** (not shown), the barrier layer **112** may extend upward to the dielectric film **104**. The plug **101** has a third width **W3** in a lateral direction **D2**. The via **102** has a via width **W4** in a lateral direction **D2**. The third width **W3** of the plug **101** may be greater than the via width **W4** of the via **102**. For example, the third width **W3** of the plug **101** is greater than two times of the via width **W4** of the via **102**.

$$\left(\frac{W3}{W4} > 2\right).$$

The longitudinal direction **D1** may be substantially perpendicular to the lateral direction **D2**. In an embodiment, the third width **W3** of the plug **101** is between 100 nm (nanometer) and 500 nm. For example, the third width **W3** of the plug **101** is about 300 nm. In an embodiment, the via width **W4** of the via **102** is between 50 nm and 250 nm. For example, the via width **W4** of the via **102** is about 100 nm.

Referring to FIG. 2, FIG. 2 illustrates a cross-sectional view of a semiconductor element **20** according to another embodiment of the present disclosure. The semiconductor element **20** is different from the semiconductor element **10** with the following description. The semiconductor element **20** further includes an isolation film **205** between the dielectric film **104** and the via **202**.

The via **202** may be in the dielectric film **104**. The via **202** is on the plug **101**. The plug **101** and the via **202** may overlap with each other in the longitudinal direction **D1**. The via **202** includes a conductive via **231** and a barrier film **232**. The barrier film **232** may be on a sidewall **231S** and a lower surface **231B** of the conductive via **231**. Part of the barrier film **232** may be between the conductive via **231** and the dielectric film **104** and between the conductive via **231** and the isolation film **205**. Another part of the barrier film **232** may be between the conductive layer **113** and the conductive via **231**. In an embodiment, the barrier film **232** directly contacts the isolation film **205** and/or the conductive via **231**. In an embodiment, the barrier film **232** directly contacts the conductive layer **113**. The conductive via **231** of the via **202**, the barrier film **232** of the via **202**, the conductive layer **113** of the plug **101**, the tungsten plug **111** of the plug **101** and the barrier layer **112** of the plug **101** are electrically connected to each other.

In this embodiment, the via **202** has a via width **W5** in a lateral direction **D2**. The third width **W3** of the plug **101** may be greater than the via width **W5** of the via **202**. For

example, the third width **W3** of the plug **101** is greater than three times of the via width **W5** of the via **202** ($W3/W5 > 3$). In an embodiment, the via width **W5** of the via **202** is between 5 nm and 90 nm. For example, the via width **W5** of the via **202** is about 25 nm.

FIGS. 3-10 schematically illustrate a method for manufacturing a semiconductor element according to an embodiment of the present disclosure.

Referring to FIG. 3, A dielectric layer **103** is provided. The dielectric layer **103** may include a dielectric material such as silicon dioxide (SiO_2), silicon nitride (SiN), silicon oxynitride (SiON), aluminum oxide (Al_2O_3), etc. An opening **341** is formed in the dielectric layer **103**. A barrier layer **112A** is formed in the opening **341**. A tungsten plug **111A** is formed in the opening **341**. The tungsten plug **111A** is formed on the barrier layer **112A**. In an embodiment, the opening **341** may be formed by applying an etching process, such as a wet etching process or a dry etching process, to remove a portion of the dielectric layer **103**. The opening **341** exposes a sidewall of the dielectric layer **103**. The barrier layer **112A** may be formed on an upper surface **103U** of the dielectric layer **103** and lining the opening **341** by a deposition process such as a chemical vapor deposition (CVD) process. A plug material layer may be formed on the barrier layer **112A** by a deposition process such as a chemical vapor deposition process. Then, portions of the barrier layer **112A** and the plug material layer on the upper surface **103U** of the dielectric layer **103** are removed by a chemical-mechanical planarization (CMP) process or other suitable etching processes, and portions of the barrier layer **112A** and the plug material layer in the opening **341** are remained. By which, the structure shown in FIG. 3 is formed. The portion of the plug material layer in the opening **341** is the tungsten plug **111A**.

Referring to FIG. 4, Portions of the tungsten plug **111A** and the barrier layer **112A** are removed to form a tungsten plug **111** and a barrier layer **112** respectively. An upper surface **111U** of the tungsten plug **111** and an end portion **112E** of the barrier layer **112** may be lower than the upper surface **103U** of the dielectric layer **103** in the longitudinal direction **D1**. The plug material layer, the tungsten plug **111A** and the tungsten plug **111** may include tungsten. The barrier layer **112A** and the barrier layer **112** may include metal barrier materials such as titanium nitride (TiN), tantalum (Ta), cobalt (Co), titanium (Ti), tantalum nitride (TaN), silicon (Si), manganese (Mn), hafnium (Hf), etc. In an embodiment, portions of the tungsten plug **111A** and the barrier layer **112A** may be removed by a chemical-mechanical planarization process or other suitable etching processes. In another embodiment (not shown), a portion of the tungsten plug **111A** may be removed to form the tungsten plug **111**, and the barrier layer **112A** is remained; the upper surface **111U** of the tungsten plug **111** is lower than the upper surface **103U** of the dielectric layer **103** in the longitudinal direction **D1**, and the barrier layer maintain the structure shown in FIG. 3.

In the longitudinal direction **D1**, the absolute value of the height difference between the upper surface **111U** of the tungsten plug **111** and the upper surface **103U** of the dielectric layer **103** may be between 300 Angstroms ($\text{\AA}$) and 2000 Angstroms ($\text{\AA}$). In the longitudinal direction **D1**, the absolute value of the height difference between the end portion **112E** of the barrier layer **112** and the upper surface **103U** of the dielectric layer **103** may be between 0 Angstroms ($\text{\AA}$) and 2000 Angstroms ($\text{\AA}$).

Referring to FIGS. 5-6, A conductive layer **113A** is formed on the tungsten plug **111** and the dielectric layer **103**.

5

A portion of the conductive layer **113A** on the upper surface **103U** of the dielectric layer **103** is removed and a conductive layer **113** shown in FIG. **6** is formed. A plug **101** is formed. The conductive layer **113** may be in the opening **341**. The conductive layer **113A** and the conductive layer **113** may include conductive materials such as titanium nitride (TiN), titanium (Ti), tantalum (Ta), tantalum nitride (TaN), etc. The conductive layer **113A/113** may directly contact the dielectric layer **103**. In an embodiment, the conductive layer **113A** may be formed on the upper surface **111U** of the tungsten plug **111**, the end portion **112E** of the barrier layer **112**, and the upper surface **103U** of the dielectric layer **103** by a deposition process such as a chemical vapor deposition process. A portion of the conductive layer **113A** on the upper surface **103U** of the dielectric layer **103** may be removed by a chemical-mechanical planarization process or other suitable etching processes. The conductive layer **113A** shown in FIG. **5** has a thickness **T1** in the longitudinal direction **D1**. The thickness **T1** of the conductive layer **113A** may be between 400 Angstroms (Å) and 2100 Angstroms (Å). For example, the thickness **T1** of the conductive layer **113A** in the longitudinal direction **D1** may be about 800 Angstroms (Å).

Referring to FIGS. **7-8**. A dielectric film **104A** is formed on the conductive layer **113** and the dielectric layer **103**. A portion of the dielectric film **104A** is removed to form a dielectric film **104** and a hole **841** in the dielectric film **104**, as shown in FIG. **8**. The bottom of the hole **841** exposes the conductive layer **113**. The hole **841** may be smaller than the opening **341**. The dielectric film **104A** and the dielectric film **104** may include dielectric materials such as silicon dioxide (SiO₂), silicon nitride (SiN), silicon oxynitride (SiON), aluminum oxide (Al₂O₃), etc. In an embodiment, the dielectric film **104A** may be formed on an upper surface **113U** of the conductive layer **113** and the upper surface **103U** of the dielectric layer **103** by a deposition process such as a chemical vapor deposition process. The hole **841** may be formed in the dielectric film **104A** by an etching process such as a wet etching process or a dry etching process. The portion of the dielectric film **104A** remained in this step is the dielectric film **104**. The dielectric film **104A/104** may have a thickness between 200 Angstroms (Å) and 2000 Angstroms (Å) in the longitudinal direction **D1**. For example, the thickness of the dielectric film **104A/104** in the longitudinal direction **D1** may be about 1000 Angstroms (Å).

Referring to FIGS. **9-10**. A barrier film **132A** is formed on the dielectric film **104** and in the hole **841**. A via material film **131A** is formed on the barrier film **132A** and in the hole **841**. Portions of the barrier film **132A** and the via material film **131A** on an upper surface **104U** of the dielectric film **104** are removed and portions of the barrier film **132A** and the via material film **131A** in the hole **841** are remained to form a barrier film **132** and a conductive via **131** shown in FIG. **10**. By which, the via **102** is formed. The portion of the via material film **131A** in the hole **841** is the conductive via **131**. The via material film **131A** and the conductive via **131** may include conductive materials such as titanium nitride (TiN), titanium (Ti), tantalum (Ta), tantalum nitride (TaN), tungsten (W), silicon (Si), Copper (Cu), Hafnium (Hf), etc. The barrier film **132A** and the barrier film **132** may include metal barrier materials such as titanium nitride (TiN), tantalum (Ta), cobalt (Co), titanium (Ti), tantalum nitride (TaN), silicon (Si), manganese (Mn), hafnium (Hf), etc. In an embodiment, the barrier film **132A** may be formed on the upper surface **104U** of the dielectric film **104** and lining the hole **841** by a deposition process such as a chemical vapor

6

deposition process. The via material film **131A** may fill the remaining space of the hole **841** and be disposed on the barrier film **132A** by a deposition process such as a chemical vapor deposition process. Portions of the barrier film **132A** and the via material film **131A** on the upper surface **104U** of the dielectric film **104** may be removed to form the via **102** by a chemical-mechanical planarization process or other suitable etching processes.

In an embodiment, a semiconductor element **10** shown in FIG. **1** may be provided through the method schematically illustrated in FIGS. **3-10**.

FIGS. **11-16** schematically illustrate a method for manufacturing a semiconductor element according to another embodiment of the present disclosure. In an embodiment, the manufacturing steps illustrated with referring to FIGS. **11-16** may be performed after the manufacturing steps illustrated with referring to FIGS. **3-8**.

Referring to FIG. **11**. An isolation material film **205A** is formed on the dielectric film **104** and in the hole **841**. For example, the isolation material film **205A** may be formed by a deposition process such as a chemical vapor deposition process. The isolation material film **205A** may include a dielectric material such as silicon dioxide (SiO₂), silicon nitride (SiN), silicon oxynitride (SiON), aluminum oxide (Al₂O₃), etc. The isolation material film **205A** may have a thickness between 500 Angstroms (Å) and 2500 Angstroms (Å) in the longitudinal direction **D1**. For example, the thickness of the isolation material film **205A** in the longitudinal direction **D1** may be about 1000 Angstroms (Å).

Referring to FIG. **12**. An etching inhibition film **206A** is formed on the isolation material film **205A**. For example, the etching inhibition film **206A** may be formed by a deposition process such as a chemical vapor deposition process. The etching inhibition film **206A** may include titanium nitride (TiN), tantalum nitride (TaN), titanium (Ti), tantalum (Ta), silicon (Si), silicon nitride (SiN), silicon dioxide (SiO₂), aluminum nitride (AlN), etc. The etching inhibition film **206A** and the isolation material film **205A** may include different materials. The etching inhibition film **206A** may have a thickness between 10 Angstroms (Å) and 100 Angstroms (Å) in the longitudinal direction **D1**. For example, the thickness of the etching inhibition film **206A** in the longitudinal direction **D1** may be about 25 Angstroms (Å).

Referring to FIG. **13**. A portion of the etching inhibition film **206A** is removed to form an etching inhibition film **206B** on a sidewall **205S** of the isolation material film **205A**. In an embodiment, the portions of the etching inhibition film **206A** on an upper surfaces **205U1** and **205U2** of the isolation material film **205A** may be removed by an etching process such as a wet etching process or a dry etching process, and a portion of the etching inhibition film **206A** on the sidewall **205S** of the isolation material film **205A** is remained to form the etching inhibition film **206B**. In an embodiment, the end portion **206BE** of the etching inhibition film **206B** may be lower than the upper surface **205U1** of the isolation material film **205A** in the longitudinal direction **D1**, which ensures that the portion of the etching inhibition film **206A** on the upper surface **205U2** of the isolation material film **205A** is removed and is helpful for the subsequent steps.

Referring to FIG. **14**. A portion of the isolation material film **205A** is removed to form an isolation film **205** on a sidewall **104S** of the dielectric film **104** and a hole **941** in the isolation film **205**. The portion of the isolation material film **205A** remained in this step is the isolation film **205**. The bottom of the hole **941** exposes the conductive layer **113**. The hole **941** may be smaller than the opening **341**. The

isolation film **205** may include a dielectric material such as silicon dioxide (SiO₂), silicon nitride (SiN), silicon oxynitride (SiON), aluminum oxide (Al₂O₃), etc. In an embodiment, the etching inhibition film **206B**, a portion of the isolation material film **205A** on the upper surface **104U** of the dielectric film **104**, and a portion of the isolation material film **205A** below the etching inhibition film **206B** are removed to expose the conductive layer **113** and the dielectric film **104**.

In another embodiment, in the step of formation of the isolation film **205**, the etching inhibition film **206B** may not be completely removed. For example, a portion of the etching inhibition film **206B**, a portion of the isolation material film **205A** on the upper surface **104U** of the dielectric film **104**, and a portion of the isolation material film **205A** below the etching inhibition film **206B** are removed to expose the conductive layer **113** and the dielectric film **104**. The etching inhibition film **206B** remained in this step is the etching inhibition film **206C**, as shown in FIG. **14A**. The etching inhibition film **206C** is on the sidewall of the isolation film **205**.

Referring to FIGS. **15-16**. After the step shown in FIG. **14**, a barrier film **232A** is formed on the dielectric film **104** and in the hole **941**. A via material film **231A** is formed on the barrier film **232A** and in the hole **941**. Portions of the barrier film **232A** and the via material film **231A** on the upper surface **104U** of the dielectric film **104** are removed and portions of the barrier film **232A** and the via material film **231A** in the hole **941** are remained to form a barrier film **232** and a conductive via **231** shown in FIG. **16**. By which, the via **202** is formed. The portion of the via material film **231A** in the hole **941** is the conductive via **231**. The via material film **231A** and the conductive via **231** may include conductive materials such as titanium nitride (TiN), titanium (Ti), tantalum (Ta), tantalum nitride (TaN), tungsten (W), silicon (Si), Copper (Cu), Hafnium (Hf), etc. The barrier film **232A** and the barrier film **232** may include metal barrier materials such as titanium nitride (TiN), tantalum (Ta), cobalt (Co), titanium (Ti), tantalum nitride (TaN), silicon (Si), manganese (Mn), hafnium (Hf), etc. In an embodiment, the barrier film **232A** may be formed on the upper surface **104U** of the dielectric film **104** and lining the hole **941** by a deposition process such as a chemical vapor deposition process. The via material film **231A** may fill the remaining space of the hole **941** and be disposed on the barrier film **232A** by a deposition process such as a chemical vapor deposition process. Portions of the barrier film **232A** and the via material film **231A** on the upper surface **104U** of the dielectric film **104** may be removed to form the via **202** by a chemical-mechanical planarization process or other suitable etching processes.

In an embodiment, a semiconductor element **20** shown in FIG. **2** may be provided through the method schematically illustrated in FIGS. **11-16**.

In a comparative example, a plug of a semiconductor element does not include a conductive layer on the tungsten plug, and the via directly lands on the tungsten plug. Since the surface of the tungsten plug is usually rough and it is difficult to reduce the roughness of the surface of the tungsten plug, the structure of the comparative example may cause the problems of profile defect of the tungsten plug and improper landing of the via on the tungsten plug. As such, the electrical performance of the semiconductor element is poor. For example, the via and the tungsten plug are electrically insulated from each other, resulting in open circuit problems. For example, at least part of the tungsten plug is not covered by the dielectric film, resulting in open circuit problems.

According to embodiments of the present disclosure, the plug of the semiconductor element includes a conductive layer between the tungsten plug and the via, and the via can land on the conductive layer. With such a configuration, the problem of poor electrical performance of the device resulting from the profile defect of the tungsten plug can be avoided, and a proper landing of the via on the plug can be ensured. The plug of the present disclosure has a desired profile. A good electrical connection between the plug and the via can be achieved. Yield and electrical performance of the semiconductor elements can be improved significantly.

It is noted that the elements and methods as described above are provided for illustration. The disclosure is not limited to the configurations and procedures disclosed above. Other embodiments with different configurations of known elements can be applicable, and the exemplified elements could be adjusted and changed based on the actual needs of the practical applications. It is, of course, noted that the configurations of figures are depicted only for demonstration, not for limitation. Thus, it is known by people skilled in the art that the related elements and layers in a semiconductor element, the shapes or positional relationship of the elements and the procedure details could be adjusted or changed according to the actual requirements and/or manufacturing steps of the practical applications.

While the disclosure has been described by way of example and in terms of the exemplary embodiment(s), it is to be understood that the disclosure is not limited thereto. On the contrary, it is intended to cover various modifications and similar arrangements and procedures, and the scope of the appended claims therefore should be accorded the broadest interpretation so as to encompass all such modifications and similar arrangements and procedures.

What is claimed is:

1. A semiconductor element, comprising:

- a plug comprising a tungsten plug and a conductive layer on the tungsten plug, wherein the tungsten plug and the conductive layer comprise different materials, the tungsten plug has a first width in a lateral direction, the conductive layer has a second width in the lateral direction, the second width is greater than or equal to the first width;
- a via on the plug and electrically connected to the plug, wherein the conductive layer is between the via and the tungsten plug, the via comprises a conductive via and a barrier film on a sidewall of the conductive via, part of the barrier film is between the conductive via and the conductive layer; and
- a dielectric layer, wherein the tungsten plug and the conductive layer are in the dielectric layer.

2. The semiconductor element according to claim 1, wherein the plug has a third width in the lateral direction, the via has a via width in the lateral direction, the third width is greater than two times of the via width.

3. The semiconductor element according to claim 2, wherein the third width is between 100 nm and 500 nm, the via width is between 50 nm and 250 nm.

4. The semiconductor element according to claim 1, wherein the plug has a third width in the lateral direction, the via has a via width in the lateral direction, the third width is greater than three times of the via width.

5. The semiconductor element according to claim 4, wherein the third width is between 100 nm and 500 nm, the via width is between 5 nm and 90 nm.

6. The semiconductor element according to claim 1, wherein the via is electrically connected to the conductive layer and the tungsten plug.

9

7. The semiconductor element according to claim 1, wherein the plug comprises a barrier layer on a sidewall of the tungsten plug, the barrier layer and the tungsten plug directly contact the conductive layer.

8. The semiconductor element according to claim 1, wherein the tungsten plug and the conductive layer are within an opening penetrating the dielectric layer along a direction, and the conductive layer is above the tungsten plug in the direction.

9. The semiconductor element according to claim 1, wherein the barrier film directly contacts the conductive layer.

10. The semiconductor element according to claim 1, wherein an upper surface of the conductive layer is lower than an upper surface of the dielectric layer.

11. The semiconductor element according to claim 1, wherein an upper surface of the conductive layer is higher than an upper surface of the dielectric layer.

12. The semiconductor element according to claim 1, further comprising a dielectric film and an isolation film, the via is in the dielectric film, the isolation film is between the dielectric film and the via.

13. The semiconductor element according to claim 12, wherein the barrier film directly contacts the conductive layer and the isolation film.

14. The semiconductor element according to claim 1, wherein a roughness of an upper surface of the conductive layer is less than a roughness of an upper surface of the tungsten plug.

15. A method for manufacturing a semiconductor element, comprising:
 forming an opening in a dielectric layer;
 forming a tungsten plug in the opening;
 forming a conductive layer above the tungsten plug, wherein the conductive layer and the tungsten plug are in the opening and in the dielectric layer; and
 forming a via on the conductive layer, wherein the via is electrically connected to the conductive layer and the

10

tungsten plug, the tungsten plug and the conductive layer comprises different materials, the via comprises a conductive via and a barrier film on a sidewall of the conductive via, part of the barrier film is between the conductive via and the conductive layer.

16. The method according to claim 15, further comprising:

forming a dielectric film on the conductive layer;
 forming a hole in the dielectric film, wherein the hole is smaller than the opening, the hole exposes the conductive layer; and
 forming the via in the hole.

17. The method according to claim 15, further comprising:

forming a dielectric film on the conductive layer;
 forming an isolation film on a sidewall of the dielectric film, wherein the isolation film is between the via and the dielectric film.

18. The method according to claim 17, further comprising:

forming a hole in the isolation film, wherein the hole is smaller than the opening and exposing the conductive layer.

19. The method according to claim 15, further comprising:

forming an isolation material film on the conductive layer;
 forming an etching inhibition film on the isolation material film, wherein the etching inhibition film and the isolation material film comprises different materials; and

removing a portion of the etching inhibition film and a portion of the isolation material film to expose the conductive layer.

20. The method according to claim 15, wherein the tungsten plug has a first width in a lateral direction, the conductive layer has a second width in the lateral direction, the second width is greater than or equal to the first width.

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